Module B - Client Server

IT NETWORK SYSTEMS ADMINISTRATION

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Introduction

You have been appointed as the system administrator for **itnsa.local** and **skill39.net**. This organization consists of Linux and Windows servers that are used to provide resources for clients within the organization.

Your role is to ensure all services are properly configured, integrated, and operational within the given time constraints. Attention to detail and time management are critical to your success in this assignment.

Credential Information

Windows & Linux

- **Username:** Administrator / root
- **Password:** Skill39!

Description of Project and Tasks

Basic Configuration

Configure **hostname**, **FQDN**, and **IP addresses** on all hosts according to the addressing table. Set the timezone to **(UTC+07:00) Bangkok, Hanoi, Jakarta** on *Windows* and **Asia/Jakarta** on *Linux*.

Active Directory

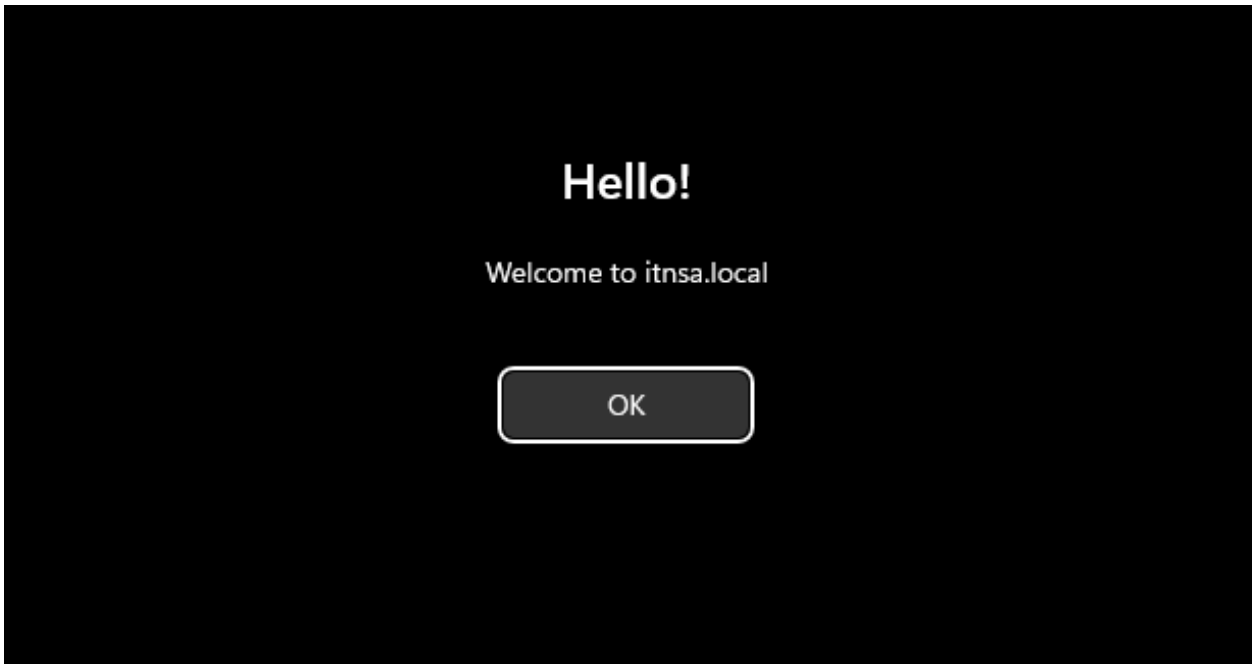
dc1.itnsa.local

1. Configure this server as the initial domain controller for a new forest named **itnsa.local**.
2. Create the following **Organizational Units (OU)** inside the domain:
   - **East**
   - **Central**
   - **West**
3. Create following **AD Groups** inside corresponding OU (*Example: **East Group** must be created inside **East** OU*) refer to the list below:
   - **East Group**
   - **Central Group**
   - **West Group**

4. Create Active Directory user accounts according to the table below. Use password **Skill39!** and make sure users do not need to change their password at the next logon.

Username	Total Accounts	Target OU
est-1 – est-5	5	East
central-1 – central-10	10	Central
wst-1 – wst-5	5	West

5. Configure group membership for all created users. Each user must be a member of the security group that corresponds with their OU (Example: ***est-1*** *assigned as a group member of* ***East Group***).
6. Create a Group Policy called **Disable Animation** that is used to disable **first sign-in animation**.
7. Create a Group Policy called **Banner** that configures a login banner with title **Hello!** And text **Welcome to itnsa.local** like example below:



DHCP Service

dc1.itnsa.local

1. Install and configure DHCP services on this server.
2. Create a new DHCP scope with the following specifications:

Setting	Value
Scope Name	ITNSA
Range	192.168.10.101 – 192.168.10.200
Subnet Mask	255.255.255.0
Excluded Address Range	192.168.10.150 – 192.168.10.160
Gateway	192.168.10.254
DNS Servers	192.168.10.1
DNS Domain Name	itnsa.local

3. Authorize this server in Active Directory as an authoritative DHCP server for the domain.

DNS Service

dc1.itnsa.local

1. Install the DNS Server role and create a Forward Lookup Zone named **itnsa.local**.
2. Create a Reverse Lookup Zone for the **192.168.10.0/24** network and create following DNS record:

Type	Record Name	Value / Target
PTR	dc1.itnsa.local	192.168.10.1
PTR	srv.itnsa.local	192.168.10.2

3. Create the following DNS records in the **itnsa.local** zone:

Type	Record Name	Value / Target
A	dc1.itnsa.local	192.168.10.1
A	srv.itnsa.local	192.168.10.2
CNAME	www.itnsa.local	dc1.itnsa.local

4. Configure a **DNS Forwarder** on this server pointing to the IP address of **lnxsrv.skill39.net (192.168.10.10)**.
5. Make sure that the **skill39.net** domain can be successfully resolved from **dc1.itnsa.local**.

lnxsrv.skill39.net

1. Install the DNS service using **Bind9**.
2. Configure this server as the primary nameserver for the **skill39.net** domain.
3. Create the following A records:

Hostname	IP Address
lnxsrv.skill39.net	192.168.10.10
srv1.skill39.net	192.168.10.11
srv2.skill39.net	192.168.10.12
srv3.skill39.net	192.168.10.13

4. Create the following CNAME records:

Alias	Target
www.skill39.net	lnxsrv.skill39.net
smb.skill39.net	lnxsrv.skill39.net
secure.skill39.net	lnxsrv.skill39.net
ftp.skill39.net	lnxsrv.skill39.net

FTP Service

lnxsrv.skill39.net

1. Install the FTP service using any package (e.g., **vsftpd** or **proftpd**)
2. Create a system user named **ftpuser** with the password **ftppass**
3. Set the FTP root directory to `/var/www/website`
4. Ensure that **ftpuser** has the necessary permissions to upload files into this directory.
5. Configure the service to **chroot** the user, ensuring they cannot navigate outside of `/var/www/website` directory.

SMB Service

lnxsrv.skill39.net

1. Install and configure the SMB service to provide shared access according to the following specifications:

Share Name	Directory Path	Access Level	Permissions
Public	/smb/public	Anyone (Guest/Anonymous)	Read-Only
Private	/smb/private	**smbuser** (Password: **smbpass**)	Read/Write

Note: You must create **smbuser** with password **smbpass** using first

Shared Folder

dc1.itnsa.local

1. Create two directory `C:\Shares\Users` and `C:\Shares\Groups` on dc1
2. Share both directory as `Users` and `Groups` shared folder
3. Allow everyone to access `Users` shared folder (read-only)
4. Only **East Group**, **Central Group** and **West Group** can access `Groups` shared folder with read/write permission.

Web Server

dc1.itnsa.local

1. Install and configure the **Internet Information Services (IIS)** web server role.
2. Stop **"Default Web Site"** on IIS; however, **do not delete or remove it**.
3. Create a new website named **www.itnsa.local** with the physical path set to `C:\www\itnsa`
4. Configure an **HTTPS** binding for the site using the certificate located at `C:\Users\Administrator\www.itnsa.local.pfx`
   Note: *You must first import the .pfx certificate into the "Local Computer" Certificate Store before you can use it on the IIS website! Use password* ***Skill39!*** *to import it!*
5. Set the content of `index.html` to **"Hello from www.itnsa.local!"**

lnxsrv.skill39.net

1. Install a web server package of your choice (e.g., **Apache2** or **Nginx**).
2. Create two virtualhost **www.skill39.net** and **secure.skill39.net** with configuration refer to table below:

Virtualhost	Root Directory	Extra Config	Index Content
www.skill39.net	/var/www/website	Enable HTTPS Basic Authentication	**Welcome to www.skill39.net!**
secure.skill39.net	/var/www/secure	Enable HTTPS only	**Secure site of skill39.net!**

3. Enable **HTTPS** connections using the SSL certificate and key files located in the `/srv/cert` directory.
4. For **secure.skill39.net** virtualhost, enable Basic Authentication using user **webuser** with password **webpass** as the user for authentication

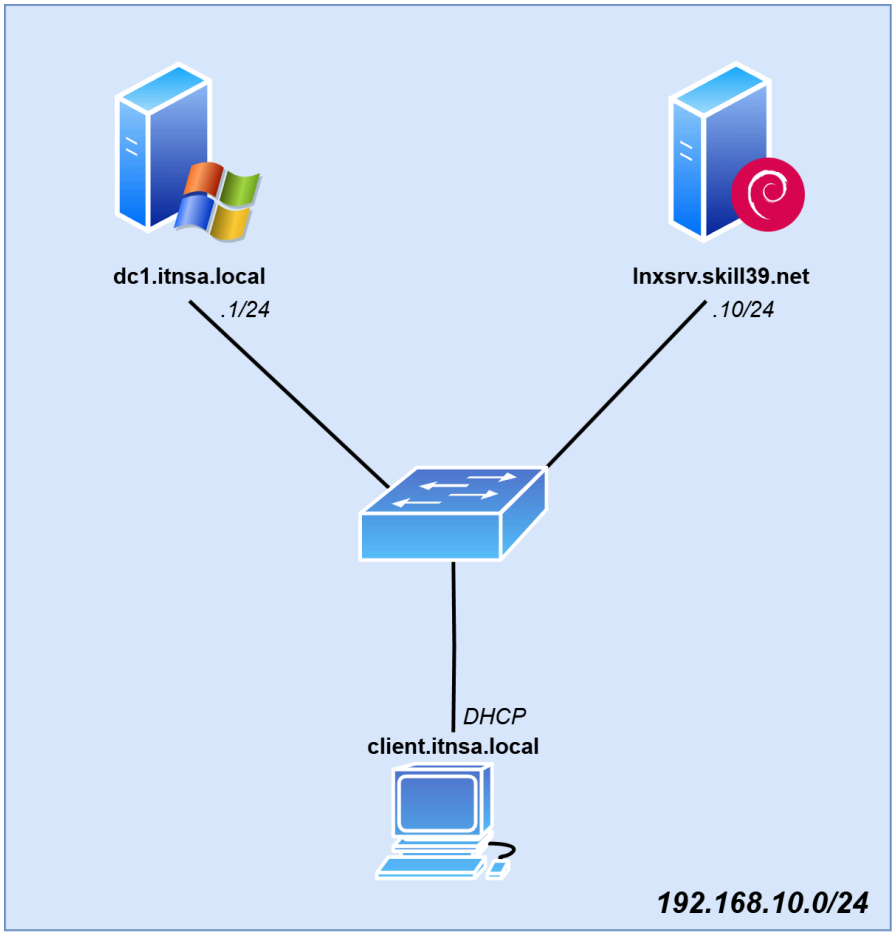
Basic Command

lnxsrv.skill39.net

1. Create directory `/opt/backup`. Then inside it, create directory `backupdir_001` until `backupdir_100` directory!
2. Backup `/smb/` directory using **tar**, then named it as `smb_backup.tar.gz`. After that, save it on `/opt/backup/backupdir_039/` directory

Appendix

Addressing Table

Hostname	FQDN	IP Address
dc1	dc1.itnsa.local	192.168.10.1/24
lnxsrv	lnxsrv.skill39.net	192.168.10.10/24
client	client.itnsa.local	DHCP

Topology